Lecture 16: Convergence in Probability, Chernoff's Bound

Jamie Haddock

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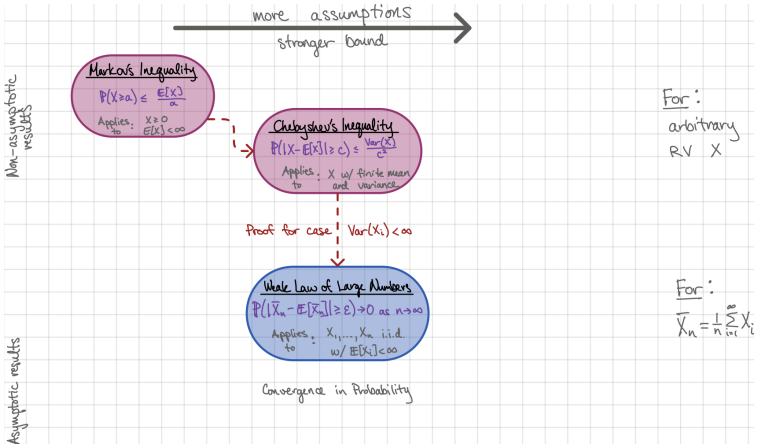
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1 Recap

We saw previously the **WLLN**, which says that $\bar{X}_n \to \mu$ in probability. Our proof used Chebyshev's inequality, and followed since $\text{Var}(\bar{X}_n) = \sigma^2/n \to 0$. However, WLLN follows even when the variance is not finite.



Today: Convergence in probability and Chernoff's bound!

2 Convergence in Probability

Reminder: A deterministic (non-random) sequence $\{a_n\}_{n=1}^{\infty}$ is said to *converge* $a_n \to a$ if for every $\varepsilon > 0$ there is n_0 such that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq n_0.$$

2.1 Convergence in Probability

We are interested in convergence of random variables now.

Definition

Let $Y_1, Y_2, \cdots$ be a sequence of random variables and let a be a real number. We say that the sequence Y_n **converges to** a **in probability** if for every $\varepsilon > 0$, we have

$$\lim_{n\to\infty} P(|Y_n - a| \geq \varepsilon) = 0.$$

WLLN interpretation: $\bar{X}_n \to \mu$ in probability.

Alternative form

Y_n converges to a in probability if for every $\varepsilon > 0$ and every $\delta > 0$ there is an n_0 such that

$$P(|Y_n - a| \geq \varepsilon) \leq \delta \quad \text{for all } n \geq n_0.$$

Here ε is the accuracy and $1 - \delta$ is the confidence.

Warning

n_0 depends on both ε and δ: more accuracy or more confidence requires larger n.

2.1.1 Example: Min of Uniform RVs

Let $U_1, \ldots, U_n$ be i.i.d. Uniform$(0, 1)$ and $Y_n = \min(U_1, \ldots, U_n)$. Where does Y_n converge?

$Y_n \to 0$ in probability: for $0 < \varepsilon < 1$,

$$P(|Y_n - 0| \geq \varepsilon) = P(Y_n \geq \varepsilon) = (1 - \varepsilon)^n \to 0.$$

Exercise

Given accuracy ε and confidence $1 - \delta$, find n_0 such that $P(|Y_n| \geq \varepsilon) \leq \delta$ for all $n \geq n_0$.

Solution

For $0 < \varepsilon < 1$ we showed $P(|Y_n| \geq \varepsilon) = (1 - \varepsilon)^n$. We need

$$(1 - \varepsilon)^n \leq \delta \iff n \ln(1 - \varepsilon) \leq \ln \delta \iff n \geq \frac{\ln \delta}{\ln(1 - \varepsilon)},$$

where the inequality flips because $\ln(1 - \varepsilon) < 0$. So take $n_0 = \left\lceil \frac{\ln \delta}{\ln(1-\varepsilon)} \right\rceil$. Since $(1 - \varepsilon)^n$ is decreasing in n, the bound then holds for all $n \geq n_0$.

3 Chernoff's Bound

Motivation. Chebyshev only gives $1/n$ decay for $P(|\bar{X}_n - \mu| \geq \varepsilon)$. For Bernoulli X_i (and many other cases), the true probability decays much faster, and thus Chebyshev is quite loose.

Can we prove a bound that decays exponentially in n? This requires more than a variance: we'll use the moment generating function!

3.1 Chernoff's Bound

Theorem (Chernoff)

Let $M_X(s) = E[e^{sX}]$. For any a and $s > 0$,

$$P(X \geq a) \leq e^{-sa} M(s).$$

Hence $P(X \geq a) \leq \min_{s>0} e^{-sa} M(s)$.

Proof. The event $\{X \geq a\}$ is the same as $\{e^{sX} \geq e^{sa}\}$. Apply Markov to the nonnegative RV e^{sX}:

$$P(e^{sX} \geq e^{sa}) \leq \frac{E[e^{sX}]}{e^{sa}} = e^{-sa} M(s). \qquad \square$$

3.1.1 Application to $\bar{X}_n$ where $X_i \sim$ Bernoulli(p)

Recall that for $X_i \sim$ Bernoulli(p), the MGF is $M_{X_i}(s) = E[e^{sX_i}] = (1-p)e^{s\cdot 0} + p\, e^{s\cdot 1} = 1 - p + pe^s$.

By Chernoff's bound and independence, we have

$$P(\bar{X}_n \geq a) = P(S_n \geq na) \leq e^{-sna} M_{S_n}(s) = \left[e^{-sa} M_{X_i}(s)\right]^n = \left[e^{-sa}(1 - p + pe^s)\right]^n.$$

3.1.2 Example: polling, revisited

Recall the polling example: we estimate the fraction p of voters supporting a candidate by $\bar{X}_n$, the fraction of n polled voters who support the candidate. Chebyshev told us that $n \geq 50{,}000$ voters suffice for the estimate to be within 0.01 of p with probability at least 0.95.

Can Chernoff's bound do better?

Overestimating p. Chernoff's bound bounds an *upper* tail, so it handles the event $\{\bar{X}_n \geq p + \epsilon\}$. By the application of Chernoff's bound to an average of Bernoulli RVs derived above with $a = p + \epsilon$, for $s > 0$ we have

$$P(\bar{X}_n \geq p + \epsilon) = P(S_n \geq n(p+\epsilon)) \leq e^{-sn(p+\epsilon)} M_{X_i}(s)^n = \left[e^{-s\epsilon} \cdot e^{-sp}\,(1 - p + pe^s)\right]^n.$$

One can show that for $p \in [0,1]$, $e^{-sp}(1 - p + pe^s) \leq e^{s^2/8}$. Then

$$P(\bar{X}_n \geq p + \epsilon) \leq e^{n(-s\epsilon + s^2/8)},$$

and optimizing with respect to s gives

$$P(\bar{X}_n \geq p + \epsilon) \leq e^{-2n\epsilon^2}.$$

Underestimating p. The lower tail $\{\bar{X}_n \leq p - \epsilon\}$ works the same way, by applying the argument to $Y_i = 1 - X_i \sim$ Bernoulli$(1-p)$:

$$P(\bar{X}_n \leq p - \epsilon) = P(\bar{Y}_n \geq (1-p) + \epsilon) \leq e^{-2n\epsilon^2}.$$

Adding the two tails gives

$$P(|\bar{X}_n - p| \geq \epsilon) \leq 2e^{-2n\epsilon^2}.$$

Now, if we want the gap $\epsilon = 0.01$ and the probability

$$P(|\bar{X}_n - p| \geq \epsilon) \leq 1 - 0.95 = 0.05,$$

then we can achieve this by choosing n such that the upper bound is less than 0.05,

$$2e^{-2n(0.01)^2} \leq 0.05 \iff n \geq \frac{\ln 40}{2(0.01)^2} \approx 18444.4,$$

or $n \geq 18{,}445$.

The exponential decay of Chernoff's bound cuts the required sample size predicted by Chebyshev's inequality by more than half!

3.1.3 Exercise

> **Exercise**
>
> Let $X \sim \text{Exp}(1)$. Use Chernoff's bound to bound $P(X \geq a)$ and compare with the exact value and with Chebyshev.

Solution

The MGF of $X \sim \text{Exp}(1)$ is

$$M(s) = E[e^{sX}] = \int_0^\infty e^{sx} e^{-x}\, dx = \frac{1}{1-s}, \qquad s < 1.$$

Chernoff's bound gives, for any $0 < s < 1$,

$$P(X \geq a) \leq e^{-sa} M(s) = \frac{e^{-sa}}{1-s}.$$

This bound is minimized at

$$s^* = 1 - \frac{1}{a},$$

where

$$P(X \geq a) \leq \frac{e^{-(a-1)}}{1/a} = ae^{1-a}.$$

Exact value. $P(X \geq a) = e^{-a}$, so the Chernoff bound is off by only the factor $e \cdot a$.

Chebyshev. For $a > 1$, $P(X \geq a) \leq P(|X - 1| \geq a - 1) \leq \dfrac{1}{(a-1)^2}$.

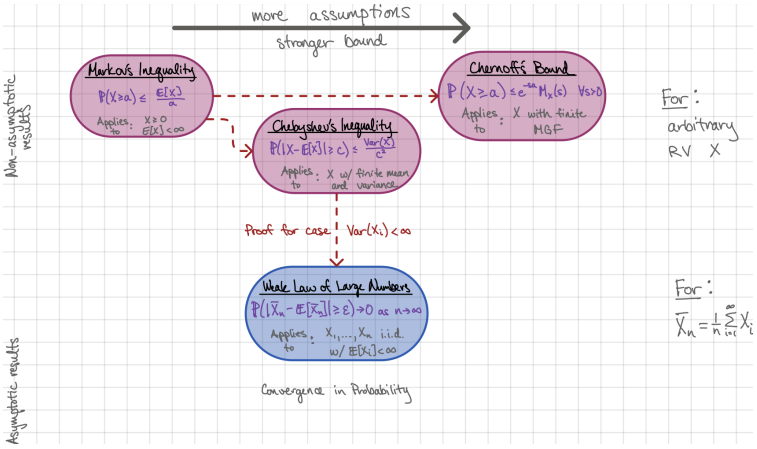
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a	Chebyshev $1/(a-1)^2$	Chernoff ae^{1-a}	Exact e^{-a}
2	1	0.736	0.135
5	0.0625	0.0916	0.00674
10	0.0123	0.00123	4.54×10^{-5}
20	0.00277	1.12×10^{-7}	2.06×10^{-9}

Chernoff has the right exponential decay, while Chebyshev decays like $1/a^2$.

3.2 Summary

Chernoff's bound requires the MGF to exist and be finite, which is stronger than assuming finite variance (Chebyshev) or mean (Markov), yields a much stronger bound!



Next time, we turn back to asymptotic convergence results for $\bar{X}_n$ and Z_n!